



Name: _____

Student Code: _____

PARTIAL FRACTION BEE

Tech-Free

Reading Time: 0 minutes

Writing time: 60 minutes

QUESTION AND ANSWER BOOK

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
6	6	63

● Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.

● Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your **name** in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

Question 1 (3 marks)

Decompose the following into partial fractions:

a. $\frac{5}{(x+1)(x+2)}$

1 mark

b. $\frac{7}{(x-1)(x+3)}$

1 mark

c. $\frac{9}{(x+5)(x+2)}$

1 mark

Question 2 (6 marks)

Decompose the following into partial fractions:

a. $\frac{5x+7}{x^2-2x+1}$

2 marks

b. $\frac{3x+8}{(x-1)(x^2+4)}$

2 marks

c. $\frac{x^2+5x+8}{x^2+2x}$

2 marks

Question 3 (9 marks)

Decompose the following into partial fractions:

a. $\frac{4x^3 + 5x^2 - 2x + 7}{(x-2)^2(x^2+1)}$

3 marks

b. $\frac{2x^3 + 5x^2 - 7x + 11}{(x-1)(x+3)}$

3 marks

c. $\frac{5x+2}{x^2-5}$

3 marks

Question 4 (7 marks)

Decompose the following into partial fractions:

a. $\frac{\sin(x)}{\cos^2(x)-1}$

2 marks

b. Find k such that: $\frac{x+k}{x^2-1} = \frac{2}{x-1} - \frac{1}{x+1}$

1 mark

c. Suppose $\frac{2x+1}{x^2+x-2} = \frac{a}{x-1} + \frac{b}{x+2} + \frac{c}{x^2+1}$, find c

1 mark

d. Decompose into partial fractions: $\frac{e^x}{e^{2x}-1}$

2 marks

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

e. Decompose into partial fractions over Complexes: $\frac{1}{x^2+1}$

1 mark

[illegible]

Question 5 (26 marks)

Evaluate the following:

a. $\sum_{r=1}^{30} \frac{3r+2}{r(r+1)(r+2)}$

3 marks

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

b. Express $f(x)$ as a sum of two rational functions with real coefficients, each having an irreducible quadratic denominator $f(x) = \frac{1}{x^4 + 1}$

3 marks

c. Given $f(x) = \frac{1}{(x-1)^2(x+2)}$, show that $f'(x) = -\frac{A}{(x-1)^3} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$

For constants A, B, C, and determine them

4 marks

d. Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$

Find constants a,b,c such that

$$(A - I)^{-1} = aI + bA + cA^2$$

4 marks

e. Show that $\frac{\sec^2(x)}{\tan^2(x)-1} = \frac{1}{2} \left(\frac{1}{\tan(x)-1} - \frac{1}{\tan(x)+1} \right)$

3 marks

f. Using:

$$\Gamma(n+1) = n\Gamma(n),$$

Show that

$$\frac{1}{\Gamma(n+2)} = \frac{1}{\Gamma(n+1)} - \frac{n}{\Gamma(n+2)}$$

Hence evaluate

$$\sum_{n=1}^N \frac{1}{\Gamma(n+2)}$$

5 marks

g. Let $\omega = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$

Given:

$$1 + \omega + \omega^2 = 0$$

Show that:

$$\frac{1}{x^3-1} = \frac{1}{3} \left(\frac{1}{x-1} + \frac{\omega}{x-\omega} + \frac{\omega^2}{x-\omega^2} \right)$$

4 marks

Question 6 (12 marks)

Simplify the following:

a. $F(x) = \frac{e^{2x}+1}{(e^x-1)^2(e^x+1)^2}$

4 marks

[illegible]

b. $\frac{\sin(2x)}{(\cos(x) - \frac{1}{2})^2 (\cos(x) + \frac{1}{2})^2}$

4 marks

[illegible]

c. $\frac{e^{3x} + 3e^{2x} + 3e^x + 1}{(e^x - 1)^3(e^x + 1)}$

4 marks

END OF PARTIAL FRACTION BEE